Vinay Ganipineni

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Professional Summary

Highly motivated and results-oriented Data Analyst with 3 years of experience in leveraging data to drive actionable insights and support strategic decision-making. Proven ability to extract, transform, and analyze large datasets using SQL, Python, and data visualization tools such as Power BI and Tableau. Passionate about data-driven problem-solving and contributing to business success through data-informed strategies.

Education

Master's in computer science | GPA 3.68/4.0 | The University of Kansas, KS, **USA**

Relevant Coursework: Data Science, Data Mining, Computer Vision, Embedded Machine Learning, Embedded Systems, Artificial Intelligence (AI), Machine Learning, Cloud Computing, Information Technology, Data Privacy & Security.

Bachelor's in computer science with specialization Big Data Analytics | GPA 3.9/4.0 | SRM University, **India**

Technical Skills:

- **Programming:** Python (NumPy, Pandas), Java, SQL,R
- **Databases:** MySQL, Postgres, Oracle, MongoDB, MS SQL Server, Snowflake.
- **Big Data & Cloud Computing:** Apache Spark, Hadoop, AWS (S3, Redshift, Glue, Lambda, DynamoDB, EC2, EMR, Quick Sight), Azure (Data Factory, Synapse Analytics, Key Vault, Azure Monitor, Logic Apps)
- **Tools**: MS Excel, Power BI, Tableau, D3.js, Databricks, GIT, Jenkin, Apache Spark, Apache Airflow.
- **Skills**: ETL (Extract Transform Load), Data modeling, Big Data, Cloud-based data storage and processing Data visualization and reporting Data profiling, validation, and reconciliation.
- **Soft Skills**: Strong written and verbal communication, incident documentation, client-facing troubleshooting
- **Machine Learning:** Supervised/Unsupervised Learning, Predictive Modeling, Regression, Classification

Professional Experience

THE UNIVERSITY OF KANSAS — **LAWRENCE**

Graduate Research Assistant (Data analyst/engineer) — **Jan 2025 – Present**

- Developed and maintained data pipelines for collecting, processing, and analyzing system performance metrics, leading to a 10% improvement in system efficiency and a 5% reduction in resource usage. Utilized Python, SQL, Apache Spark, and Azure services for real-time data transformation and monitoring.
- Designed and implemented data-driven algorithms for computational resource optimization, resulting in a 15% reduction in processing time for key data analysis tasks.
- Collaborated with stakeholders to define key performance indicators (KPIs) and develop dashboards using Power BI to track and monitor system performance, enabling data-informed decision-making.
- Supported cloud-based compute environments across Azure, assisting with virtual machine setup, storage configuration, and infrastructure troubleshooting, ensuring optimal performance and scalability.
- Leveraged Azure tools including Key Vault, Logic Apps, Synapse, and Monitor for secure, scalable cloud operations, meeting compliance requirements and minimizing security risks.

Research Fellowship (USA) — **Sep 2023 – Dec 2024**

- Designed interactive Power BI dashboards for real-time sales performance tracking, integrating Azure Logic Apps to trigger automated notifications, leading to a 20% increase in sales team responsiveness.
- Built and optimized ETL pipelines using Apache Spark and Databricks, ensuring efficient data ingestion, transformation, and storage, resulting in a 40% reduction in data processing time.
- Implemented real-time data streaming with Kafka, enabling immediate analysis of key sales trends and providing actionable insights for sales managers.
- Utilized Azure cloud services (Key Vault, Azure Monitor, Mapping Data Flows, Medallion Architecture) for secure data management, monitoring, and transformation workflows, ensuring data integrity and security.
- Orchestrated ETL workflows using Apache Airflow, automating data processing and ensuring seamless execution of data pipelines, improving efficiency and reducing manual effort.

Product and Engineering (Full stack developer) , High Radius (Ind) — **Jan 2022 – Apr 2022**

- Designed ETL pipelines to process financial transactions, enabling accurate invoice payment predictions with ML models, leading to a 10% improvement in invoice payment accuracy.
- Built and optimized SQL databases for efficient data storage and real-time analytics, supporting data-driven decision-making and providing timely information for key stakeholders.

- Integrated big data tools (Spark, Kafka) for streaming financial data processing, enabling real-time analysis of financial trends and identification of potential anomalies.
- Automated data validation and cleansing, improving data accuracy and reducing discrepancies by 25%, leading to more reliable financial reports.
- Developed APIs and microservices for seamless data flow between front-end apps and cloud storage, ensuring efficient data access and data integration across different systems.

Projects

Customer Churn Prediction for a Subscription-Based Service

- Obtain customer data from the company's database, including subscription history, demographics, account activity, payment information, and interactions with customer support. Cleanse and prepare the data, handling missing values, outliers, and inconsistencies.
- Engineer relevant features from the raw data, such as average viewing time, number of songs streamed, subscription duration, and customer engagement metrics.
- Choose an appropriate machine learning model for churn prediction, such as logistic regression, random forest, or support vector machine. Split the data into training, validation, and testing sets.
- Train the model on the training data, adjusting hyperparameters to optimize model performance. Evaluate the trained model using appropriate metrics like accuracy, precision, recall, and F1 score.
- Analyze the model's predictions to identify key factors contributing to churn.
- Create interactive dashboards using Power BI or Tableau to visualize churn trends, key factors influencing churn, and the effectiveness of retention strategies. Present your findings to stakeholders with clear, concise reports.

Sales Performance Analysis and Forecasting for a Retail Company

- Gather sales data from various sources, including POS systems, online sales platforms, and customer databases and Explore the data to identify trends, seasonality, and patterns using time series analysis techniques.
- Visualize sales data using charts and graphs to gain initial insights. Engineer relevant features from the sales data, such as product category, price, promotion type, customer demographics, and seasonality indicators.
- Choose a suitable time series forecasting model, such as ARIMA, Prophet, or LSTM.
- Train the chosen model using the historical sales data, optimizing hyperparameters to improve model performance.
- Evaluate the model's accuracy using forecasting metrics like mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE).
- Analyze the forecasted sales trends to identify potential growth opportunities, and the impact of different promotional strategies.
- Provide recommendations to the sales and marketing teams based on the forecasted sales data, helping them make data-driven decisions for inventory management, pricing, and promotional campaigns.

Website Traffic Analysis and Optimization for a Travel Website

- Collect web analytics data from tools like Google Analytics, including page views, user sessions, bounce rates, conversion rates, and user demographics.
- Explore the data to identify initial patterns and areas for improvement. Analyze user behavior within each segment to understand their motivations, pain points, and conversion paths.
- Design and implement A/B tests to evaluate the effectiveness of different website changes (e.g., button placement, content updates, page design) on key metrics like conversion rates.
- Track the results of A/B tests and identify the changes that lead to the most significant improvements.
- Create dashboards using Power BI or Tableau to visualize website traffic trends, user behavior patterns, A/B test results, and the impact of optimization efforts. Present your findings to the website development and marketing teams, providing clear recommendations for further optimization.

Certifications:

- AWS Certified Data Engineer – Associates| Issued Jan 2025
- Microsoft Certified Data Analysis by Microsoft and LinkedIn
- Paper published in International Conference on Computer Science (ICSS) 2023
- Machine Learning & Big Data Using Python and Data Science with R Language Machine Learning & Big Data Using Python and Data Science with R Language Prag Robotics| Issued Apr 2022